Coxeter groups in Lean

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- Learn how to work with Coxeter groups in Lean

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- Show examples of definitions and theorems

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- Show examples of definitions and theorems
- Some potential projects

What is a Coxeter group?

Let S be a set.

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Definition

A *Coxeter matrix* is a matrix $m : S \times S \to \mathbb{N}_{\geq 1} \cup \{\infty\}$, such that

$$m(s, s') = m(s', s)$$
$$m(s, s') = 1 \iff s = s'$$

for all $s, s' \in S$.

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m determines a *Coxeter group* W, with generators S and relations

$$(ss')^{m(s,s')} = e$$

for $s, s' \in S$ with $m(s, s') \neq \infty$.

Coxeter groups in `mathlib`

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Definition in mathlib:

```
structure CoxeterMatrix (B : Type*) where
  M : Matrix B B ℕ
  isSymm : M.IsSymm
  diagonal i : M i i = 1
  off_diagonal i i' : i ≠ i' → M i i' ≠ 1
```

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```
structure CoxeterSystem {B : Type*} (M : CoxeterMatrix B) (W :
    Type*) [Group W] where
    mulEquiv : W ≃* M.Group
```

Some definitions in mathlib

```
import Mathlib.GroupTheory.Coxeter.Basic
import ...

variable {B : Type*} {W : Type*} [Group W] {M : CoxeterMatrix B}
    (cs : CoxeterSystem M W)
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- `cs.simple : B → W`
- `cs.wordProd : List B → W`
- `cs.length : W → ℕ`
- `cs.leftInvSeq : List B → List W`

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theorem getElem_leftInvSeq (ω : List B) (j : ℕ)
  (h : j < ω.length) :
  (cs.leftInvSeq ω)[j]
  = cs.wordProd (List.take j ω) * cs.simple ω[j] * (cs.wordProd
    (List.take j ω))⁻¹
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theorem IsReduced.nodup_leftInvSeq {ω : List B}
  (rω : cs.IsReduced ω) : (cs.leftInvSeq ω).Nodup
```

Formalizing more stuff

We'll follow the textbook *Combinatorics of Coxeter Groups* by Björner–Brenti.

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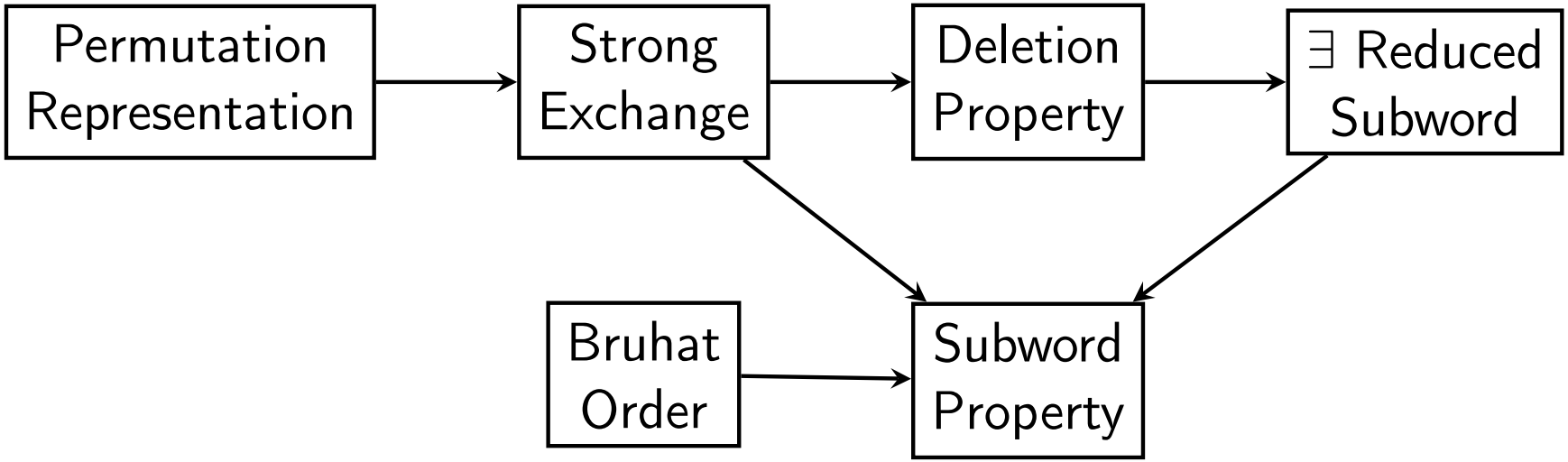

Packaging things together

Define

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class CoxeterGroup (W : Type*) extends Group W where
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Instead of writing

```
variable {B : Type*} {W : Type*} [Group W] {M : CoxeterMatrix B}
    (cs : CoxeterSystem M W)
```

we can now write instead

```
variable {W : Type*} [CoxeterGroup W]
```

Permutation representation

Define the set of reflections for W (usually denoted T):

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def ReflectionSet (W : Type*) [CoxeterGroup W] :=
  {t : W // cs.IsReflection t}
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(Omitted.)

Permutation representation

```
def permRep : W →* Equiv.Perm (RootSet W)

def η (w t : W) : ZMod 2 :=
  count t (cs.leftInvSeq (choose (cs.wordProd_surjective w)))

theorem permRep_eq (w : W) (r : RootSet W) :
  permRep w r
  = ⟨⟨MulAut.conj w r.1.val, r.1.prop.conj _⟩, r.2 + η w⁻¹
    r.1.val⟩
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In usual notation:

- $\eta(w, t)$ is the number of occurrences mod 2 of t in $s_1, s_1 s_2 s_1, s_1 s_2 s_3 s_2 s_1, \ldots$ for some choice of word $w = s_1 s_2 \ldots s_k$.
- $\pi_w(t, \varepsilon) = (wtw^{-1}, \varepsilon\eta(w^{-1}, t))$

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η is independent of the choice of word:

```
theorem eta_spec (ω : List (B W)) (t : W) :
  η (cs.wordProd ω) t = count t (cs.leftInvSeq ω)
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The proof of strong exchange uses

```
theorem eta_eq_one_iff {t w : W} :
  η w t = 1 ↔ cs.IsLeftInversion w t
```

Strong exchange & deletion property

```
theorem strong_exchange {ω : List (B W)} {t : W}
  (h : cs.IsLeftInversion (cs.wordProd ω) t) :
  ∃ i < ω.length, t * cs.wordProd ω = cs.wordProd (ω.eraseIdx i)
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theorem deletion_property {ω : List (B W)}
  (hω : ¬ cs.IsReduced ω) :
  ∃ i j, i < j ∧ j < ω.length ∧ cs.wordProd ω = cs.wordProd
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theorem exists_reduced_subword (ω : List (B W)) :
  ∃ (ω' : List (B W)), ω' <+ ω ∧ cs.IsReduced ω' ∧
    cs.wordProd ω = cs.wordProd ω'
```

Here <+ means 'is a sublist of'.

Defining the Bruhat order

Definition

For $u, w \in W$, $u \leq w$ in Bruhat order means that there is a chain

$$u = u_0 \to u_1 \to \ldots \to u_{k-1} \to u_k = w$$

where for each i, $u_{i+1} u_i^{-1}$ is a reflection and $\ell(u_i) < \ell(u_{i+1})$.

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```
inductive le : W → W → Prop
  | rfl (u : W) : le u u
  | step (u v w : W) (h1 : le u v)
      (h2 : cs.IsReflection (w * v⁻¹))
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The following lets us write `u ≤ w` instead of `le u w`:

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Note: `le` depends on `cs`. This is why it is useful to have `[CoxeterGroup W]`, so that Lean knows which `CoxeterSystem` to use when we write `u ≤ w`.

Subword property

For convenience, we define

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Finally ...

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theorem subword_property (u : W) {w : W} (ω : ReducedWord w) :
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```

This lets us prove that Bruhat intervals are finite:

```
theorem finite_Icc (u w : W) : Finite (Set.Icc u w)

instance : LocallyFiniteOrder W

theorem card_Icc_le (u w : W) :
  Finset.card (Finset.Icc u w) ≤ 2 ^ cs.length w
```

Github page

The page for this project can be found at

https://github.com/mlyean/lean-coxeter

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